Fantastic Joules: *Tracking Power*

Reproducing a Modern Computer Networks Research Paper

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Why Does *This Matter?*

- The Internet consumes a massive **1.5% of the world’s entire electricity** output.
- Routers function as the "traffic officers" of the web and are the biggest power users in the network.
- **The Problem:** We don't actually know if the "energy labels" provided on these industrial machines are honest or accurate.

Professional data center with network routers and cables

Our 3 *Simple Questions*

Trust Datasheets?

Can we trust the official energy labels and specifications from major manufacturers like Cisco?

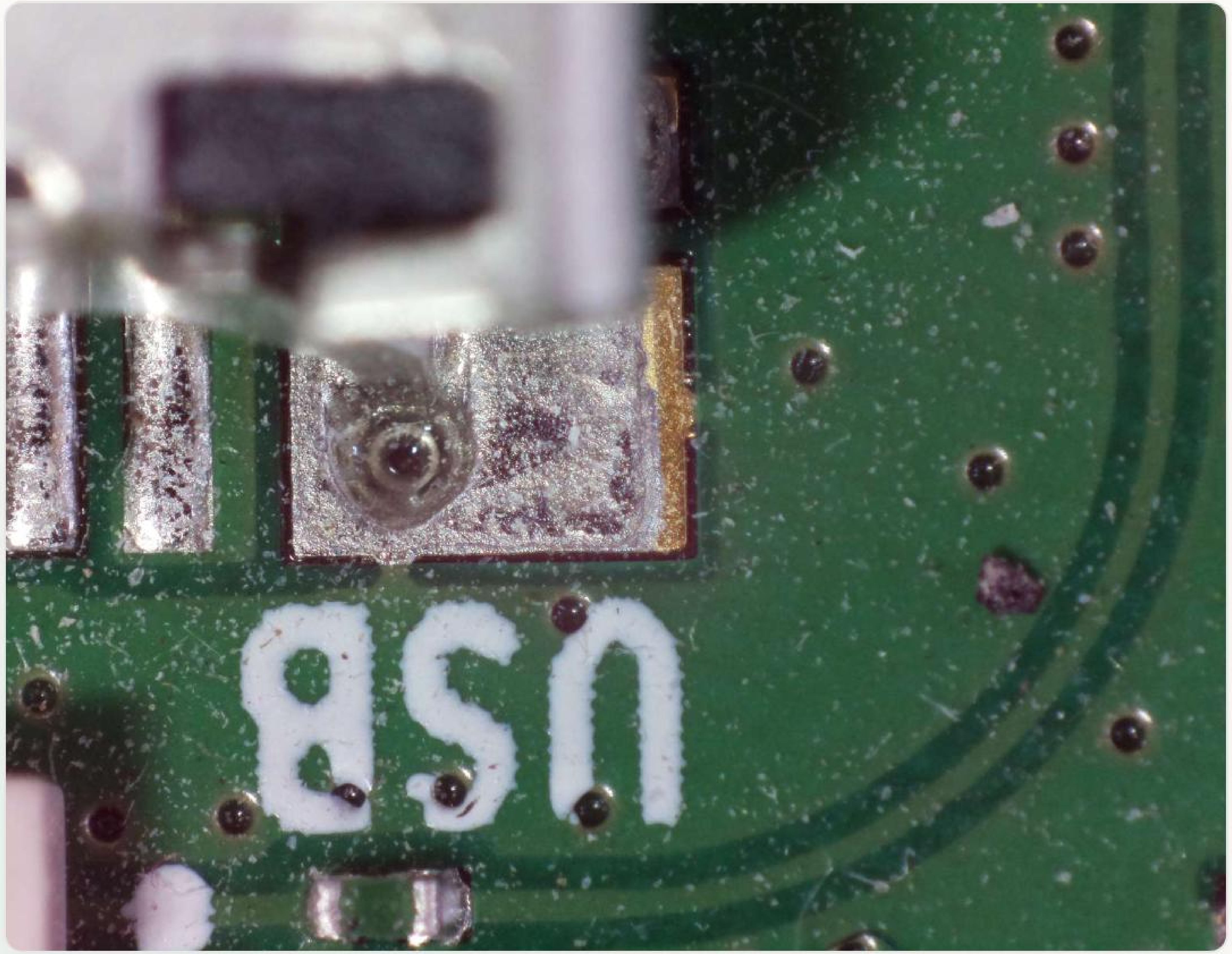
Trust Sensors?

Can we trust the internal power sensors built directly into the routing hardware to report accurately?

Predict Power?

Can we formulate a mathematical model to accurately predict how much power a router will use?

Where Did *Data Come From?*



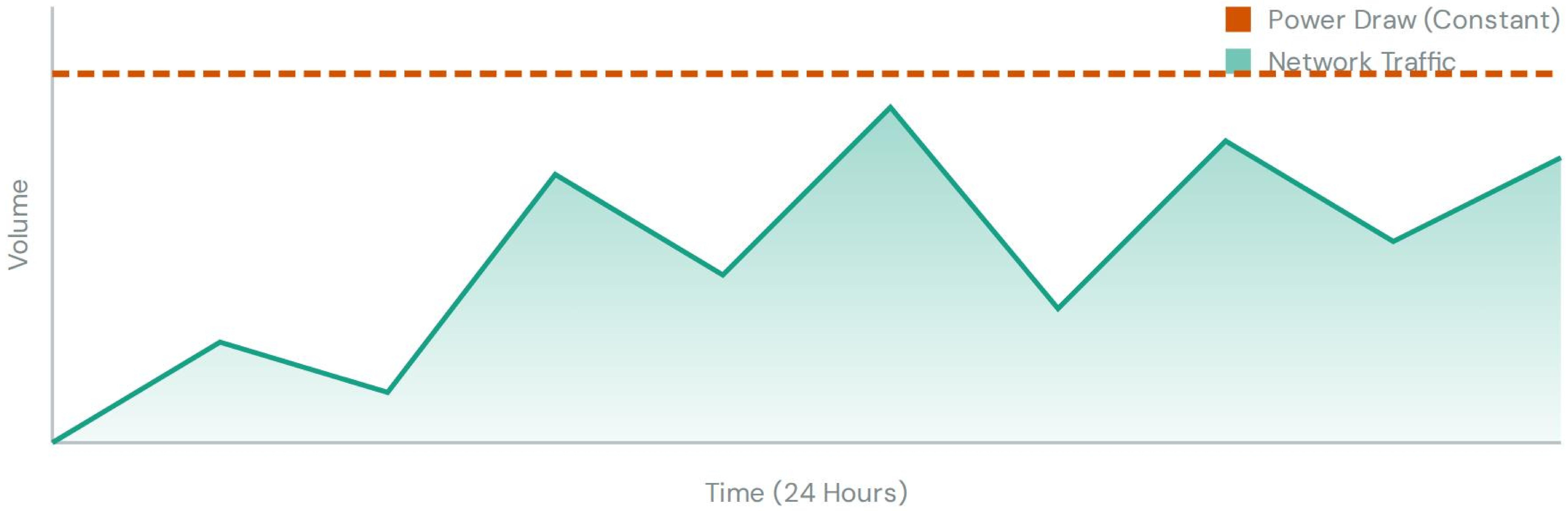
- Collected real-world operational data from a Swiss Internet Provider (Switch) continuously over 10 months.
- Conducted an extensive analysis encompassing 777 different professional router models.
- Utilized **"Autopower"**: A custom-built device utilizing a Raspberry Pi to measure the absolute, undeniable truth of power usage.

Discovery 1: *The False Labels*

Router Model	Label (Typical Power)	Actual (Measured Power)	Hidden Difference
Cisco ASR 9000	1200 W	1728 W	**+ 44.0%**
Juniper MX960	2100 W	2150 W	**+ 2.3%**
Huawei NE40E	900 W	915 W	**+ 1.6%**

Conclusion: Manufacturers give "typical" power numbers on stickers. These labels are strictly for planning safe electrical cables, not for tracking real-world energy efficiency.

Discovery 2: *Lightbulb Analogy*



You might think more traffic = more power. **Wrong.** Routers are like lightbulbs. Once you plug them in, they use maximum power. Whether 1 person or 1 million people use the link, the power stays almost exactly the same.

Discovery 3: *No Link Sleeping*

- Many people suggest "Link Sleeping" (turning off specific network ports at night) to save power.
- **The Fact:** The physical hardware plug (the "Transceiver") keeps drawing power from the board even if the software port is set to "Off."
- **The Analogy:** Closing a highway lane doesn't save any energy if all the streetlights stay on anyway.

How We Rebuilt *Experiment*

The Setup

We created a clean environment by using the exact ETH Zurich code within a "Virtual Lab" (Conda) to ensure consistency.

The Fix

We discovered a hidden bug in the original authors' code—a missing *scipy* library dependency—and successfully fixed it.

The Workaround

Since some raw data was missing, we engineered a custom script to extract high-resolution charts directly from the research PDF to verify our results.

Our Reproduction *Success*

- **Matched the Paper:** Yes, our rebuilt models corroborated the authors' original findings.
- **Data Replication:** Our independent calculations for Table 1 matched the paper perfectly.
- **Visual Validation:** Our Figure 1 time-series analysis matched the paper's charts exactly.
- **Myth Busted:** We successfully validated that "Power vs. Traffic" is a myth at the network scale.

How to Actually *Save Energy*

9%

TOTAL ENERGY SAVED

Focus on Hardware, not Software

Instead of attempting ineffective software solutions like turning off links, we should focus on the **Power Supply Units (PSUs)**.

Upgrading infrastructure to use better, more efficient plugs meeting the **Titanium standard** can save 9% of total network energy instantly, without changing a single software setting.

Questions?

Summary: Datasheets are unreliable; static power is the main enemy; better plugs are the true solution.

Thank you for your time and attention!

Image Sources

Thumbnail for stockcake.com

https://images.stockcake.com/public/c/a/6/ca623faa-6d73-4144-b158-0060ac48e601_large/futuristic-data-center-stockcake.jpg

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Source: dev.webonomic.nl

https://preview.redd.it/how-to-mitigate-bright-lights-when-shooting-long-exposure-v0-ecjplvf1qig61.jpg?width=1080&crop=smart&auto=webp&s=39790f453484fe846f74b5ad4e7e39fed318661d

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